

Applying Explainability Methods to Financial Models

Objective

State-of-the-art financial forecasting models often exhibit weak predictive performance, with classification metrics around 50% and similar limitations in return forecasting. Such instability propagates to portfolio allocation strategies, such as long-only top-k, where small prediction changes can significantly alter portfolio composition and risk. The project aims to develop a framework to analyze how input features influence model predictions and how this impact translates into allocation decisions.

A key concept in this context is the **investment universe**, which represents the set of tradable assets (e.g., equities within a given index, sector, or market) from which the model selects. The choice of universe is crucial, as it defines the opportunity set and directly impacts both model training and portfolio outcomes. Different universes (e.g., large-cap vs small-cap stocks) exhibit different statistical properties, liquidity profiles, and levels of noise.

The entire modeling pipeline typically starts from **OHLCV data** (Open, High, Low, Close, Volume). From these raw time series, a wide range of features is engineered, such as returns, volatility measures, momentum indicators, and volume-based signals, which serve as inputs to machine learning models.

Step 1

The first step is to identify a set of comparable models that use the same data and experimental setup. These models may be selected from FINBENCH, or additional architectures can be integrated if needed. The goal is to create groups of models that are directly comparable in terms of input data and evaluation protocol.

Step 2

The selected models will be trained using FINBENCH (or other new models) in order to obtain checkpoints. These trained models will then be used to apply explainability techniques such as SHAP (and LIME) or other time-series-specific techniques. The objective is to identify which input features have the greatest impact on model predictions. This analysis will allow us to:

- Rank features by importance for each model
- Compare feature importance across models
- Study whether certain features consistently dominate predictions

This step provides a quantitative understanding of how models use the available information.

Step 3

After obtaining feature attributions, a qualitative analysis will assess whether models learn similar or different patterns. If models consistently prioritize the same features, this may suggest that prediction instability stems from limited or noisy signals in the data. Conversely, if feature importance varies significantly across models, architectural differences may play a

key role. This step aims to determine whether instability originates primarily from the data or from the model design.

Step 4 (Optional)

Counterfactual explanation techniques will be used to assess the stability of predictions in portfolio allocation tasks. The objective is to measure how much inputs or predictions must change to alter portfolio composition. Since financial features are often derived and highly correlated, realistic constraints must be imposed when generating counterfactuals. This analysis will help quantify the robustness of portfolio decisions to small prediction changes.

Conclusion

The goal is to understand how data influences predictions and allocation decisions, identify the main drivers of instability, and determine whether they originate from the data or the model design. The expected outcome is a graphical interface for loading models, visualizing feature attributions, and comparing explanations across architectures.

References

- Computational Intelligence and Financial Markets: A Survey and Future Directions - Expert Systems With Applications 2016:
<https://www.sciencedirect.com/science/article/pii/S095741741630029X>
- XAI in Portfolio Management - literature review: [paper](#).
- LIME: [paper](#), SHAP: [paper](#)
- An explainable deep learning approach for stock market trend prediction -
<https://www.sciencedirect.com/science/article/pii/S2405844024161269>
- Regression models based on Alphas:
 - MATCC - CIKM 2024: <https://dl.acm.org/doi/10.1145/3627673.3679715>
 - MASTER - AAAI 2024: <https://ojs.aaai.org/index.php/AAAI/article/view/27767>
 - HIST - arxiv 2022: <https://arxiv.org/pdf/2110.13716>
 - DiscoverPLF - TKDE 2023:
<https://ieeexplore.ieee.org/stamp/stamp.jsp?tp=&arnumber=10324313>
 - FactorVAE - AAAI 2022:
<https://ojs.aaai.org/index.php/AAAI/article/view/20369>
 - FinFormer - IEEE International Conference on Big Data 2023:
<https://ieeexplore.ieee.org/stamp/stamp.jsp?tp=&arnumber=10386751>
- Qlib - arxiv 2020: <https://arxiv.org/pdf/2009.11189> (*Questo è il paper che la prima volta ha introdotto il concetto di Alpha - L'implementazione corretta la trovi comunque nella repo Finbench nella directory Evaluation*)

Link utili:

- Repo Finbench: <https://github.com/softlab-unimore/finbench>

- Tutti i dati li trovi qua su Drive nella cartella **Data**. Quelli che principalmente ti serviranno sono gli Alpha158 e gli Alpha360 sugli universi più piccoli (NASDAQ100, DJI, SX5E). Questi indicatori si ricavano tutti dagli OHLCV.

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What to do:

Prima parte:

1. Leggere la survey: è un po' vecchiotta, ma da una buona panoramica su come è la pipeline usata in questo contesto.
2. Leggere i paper che parlano dei vari modelli di regressione. L'obiettivo è comprendere il task (cosa si sta predicendo) e i dati che sono usati (+ normalizzazioni applicate)
3. Leggere i paper riguardanti XAI. *Questa è la parte più di ricerca della tesi. Una volta che hai chiaro il task e come funzionano i modelli, se riesci bisognerebbe cercare in letteratura tecniche specifiche per time series forecasting e se si ci sono per stock trend forecasting.*
4. Implementare con i modelli un framework che permetta di applicare i modelli di XAI e salvare risultati ed eventuali metriche di valutazione.
5. Test: li faremo su diversi periodi usando una rolling window. Serve capire se ci sono periodi particolari che sono soggetti a maggiore volatilità e così via.

Seconda parte:

1. Test aggiuntivi che definiremo in seguito in base ai risultati ottenuti. Molto probabilmente riguarderanno tecniche di modifiche dei dati per capire quanto sensibili sono i modelli.